IP Reuse In SoC

Due to the reuse of components in an SoC, the design mechanics for the product undergoes a major transition from an application-specific IP (ASIP) design to a block-based design (BBD), and then finally to a platform-based plug and play SoC design.

For a BBD, the electronic system is divided into different mechanisms and these categories are mapped into available IP components. Their placements are retained, and the routing, timing etc, are all re-optimised to aid in the overall optimisation of the entire system. Following this, the BBD is transformed to a platform-based (PBD) design. This PBD allows support for Hard IP or Hardware IP.

Overall, reused IPs increase productivity. However, certain design challenges often arise, which the SoC makers have to tackle and counter independently. These challenges include the use of heterogeneous device technologies, issues related to testing, signal integrity, voltage maintenance and time regulation. Moreover, SoCs are application specific and, therefore, the application of the chip defines the IP. To meet the need of a particular application, these IPs are frequently redesigned.

Hardware and software development is required for an IP-based SoC. IPs which closely resemble the specifications needed to work with a particular app are selected and redesigned to support the designated application. These redesigned IPs transform into Virtual Components, or VC.

IP Infringements

An SoC company has to overhaul the design of an IP extensively to change it into VCs. However, due to constraints in time and effort, the company may hire outside designers and external tools. Often, SoC developers are unable to afford their own fabrication facilities, which compels them to seek outside help. However, the inclusion of third-parties is a cause for concern, especially in the context of maintaining the right of access for the IP.

The SoC developer may have purchased the IP from a legal vendor; however, if the design aspects of the IP are leaked by the external members working on the SoC, it results in massive revenue loss for both the developer and the IP vendor.

The following points show how an IP could be violated in these circumstances:

1. The first and foremost concern for IP creators is that the SoC company itself may acquire the IP through illegal means. The company may